

assignment1 - Intro electrostatics - coulombs 1D

 This is a preview of the published version of the quiz

Started: Mar 15 at 3:07pm

Quiz Instructions

First study the unit electrostatics in your textbook (introduction and coulombs law) and Review adding vectors and Scientific notations. (lots of tutorials online on how to add vectors algebraically. .

Idially: join a study group. A subgroup of the whastapp group.

2 attempts. I keep the last grade.

Do not look at the hints (use a post-it to hide them). Remember to convert .

There are tutorials for questions 1 and 2 in my dropbox. (try first without watching).

Work with a study group. Make new friends

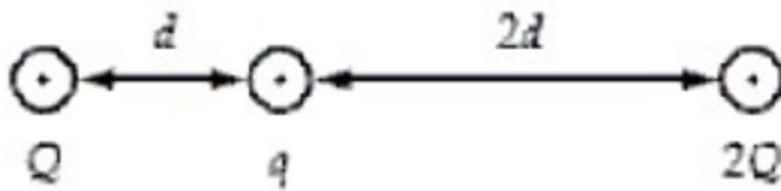
Terameter	1,000,000,000,000 m	10^{12} m	1 Tm	Tera
Gigameter	1,000,000,000 m	10^9 m	1 Gm	Giga
Megameter	1,000,000 m	10^6 m	1 Mm	Mega
Kilometer	1,000 m	10^3 m	1 Km	Kilo
meter	1 m	100 m	1 m	
decimeter	0.1 m	10^{-1} m	1 dm	deci
centimeter	0.01 m	10^{-2} m	1 cm	centi
millimeter	0.001 m	10^{-3} m	1 mm	milli
micrometer	0.000001 m	10^{-6} m	1 μ m	micro
nanometer	0.000000001 m	10^{-9} m	1 nm	nano
picometer	0.000000000001 m	10^{-12} m	1 pm	pico
femtometer	0.000000000000001 m	10^{-15} m	1 fm	femto



Question 1 4.4 pts

In the figure, if $Q = 30 \mu\text{C}$, $q = 5.0 \mu\text{C}$, and $d = 30 \text{ cm}$, what is the magnitude of the electrostatic force on q

hint: I made a video for this one. See dropbox


☐

7.5 N

☐

23N

☐

zero

☐

15N

☒

Question 2 4.4 pts

A particle ($m = 50 \text{ g}$, $q = 5.0 \mu\text{C}$) is released from rest when it is 50 cm from a second particle ($Q = -20 \mu\text{C}$). Determine the magnitude of the initial acceleration of the 50-g particle.

hint: I made a video for this one. See dropbox

☐

72 m/s^2

☐

54 m/s^2



65 m/s²



36 m/s²



Question 3 4.4 pts

To become a negative ion, an atom must



lose a proton.



gain an electron.



gain a proton.



lose an electron.



Question 4 4.4 pts

To become a positive ion, an atom must



lose an electron.



gain a proton.



lose a proton.



gain an electron.



Question 5 4.4 pts

A positive ion has more



protons than electrons.



electrons than neutrons.



neutrons than protons.



protons than neutrons.



electrons than protons.



Question 6 4.4 pts

Two charges that are separated by one meter exert 1-N forces on each other. If the charges are pushed together so the separation is 25 centimeters, the force on each charge will be

hint: convert cm to m first

hint: you can do it the fast way. the distance is changed from 1 m to 0.25m . so the distance is divided by4. Inverse square law. What happens to the force?

☐

2 N.

☐

16 N.

☐

8 N.

☐

1 N.

☐

4 N.



Question 7 4.4 pts

Two charges that are separated by one meter exert 1-N forces on each other. If the magnitude of each charge is doubled, the force on each charge is

☐

2 N.

☐

4 N.

☐

1 N.

☐

8 N.

☐

none of the above



Question 8 4.4 pts

Rub electrons from your hair with a comb and the comb becomes

hint: the comb is like a balloon

☐

positively charged.

☐

discharged.

☐

negatively charged.



Question 9 4.4 pts

Two charged particles repel each other with a force F . If the charge of one of the particles is doubled and the distance between them is also doubled, then the force will be

☐

F .

☐

$F/4$.

☐

$2 F$.

☐

$F/2$.

☐

none of the above



Question 10 4.4 pts

When the distance between two charges is halved, the electrical force between them

☐

doubles.

☐

is reduced by $1/4$.

☐

halves.

☐

quadruples.

☐

none of the above



Question 11 4.4 pts

Particle A has twice the charge of nearby particle B. Compared to the force on Particle A, the force on Particle B is

hint: Newton's third law. For every action there is an equal and opposite reaction.

☐

four times as much.

☐

the same.

☐

half as much.

☐

twice as much.

☐

none of the above



Question 12 4.4 pts

The electrical force between electric charges depends only on their

☐

magnitude.

☐

separation distance.

☐

both of these

☐

neither of these



Question 13 4.4 pts

A piece of plastic has a net charge of $+2.00 \mu\text{C}$. How many more protons than electrons does this piece of plastic have? ($e = 1.60 \times 10^{-19} \text{ C}$)

hint: $Q = N e$ (N is the number of electrons that left = number of protons not paired)

☐

2.50×10^{13}

☐

2.50×10^{19}

☐

1.25×10^{19}

☐

1.25×10^{13}



Question 14 4.4 pts

A common naturally-polarized bit of matter is

see slide distribution of charges where I go through applications in the real world.

☐

a hydrogen atom

☐

an electron.

☐

a water molecule.

☐

all of the above

☐

none of the above



Question 15 4.4 pts

If you rub an inflated balloon against your hair and place the balloon against the wall it will stick to the wall, illustrating

hint: see slide. As a charged rod is brought near a wall. Same thing when a balloon sticks to the wall.

☐

voltage.

☐

conduction and insulation.

☐

Coulomb's law.

☐

charge polarization.

☐

Question 16 4.4 pts

A positive charge and a negative charge held a certain distance apart are released. As they move, the force on each particle

hint: positive and negative, do they attract or repel? and as the distance changes, what happens to the force (inverse square law)

☐

decreases.

☐

increases.

☐

stays the same.

☐

Question 17 4.4 pts

The electrical force between charges is strongest when the charges are

☐

close together.

☐

far apart.

☐

either of these

☐

need more information

☐

Question 18 3 pts

A negatively-charged rod is held near an aluminum can that rests on a dry wood table. If you momentarily touch the opposite side of the can with your finger, the can becomes

hint: see slide. It is called charging by induction. (you ground one side)

☐

completely discharged.

☐

positively charged.

☐

partially discharged.

☐

negatively charged.

☐

none of the above



Question 19 3 pts

Two protons attract each other gravitationally and repel each other electrically. The stronger of these two forces is

☐

electrical.

☐

gravitation.

☐

neither of these



Question 20 4.4 pts

A 1.0-C point charge is 15 m from a second point charge, and the electric force on one of them due to the other is 1.0 N. What is the magnitude of the second charge?

$$k=9 \cdot 10^9$$

hint: $1 = k q_1 q_2 / (15^2)$ with $q_1=1$ solve for q_2

☐

10 nC

☐

25 nC

☐

1.0 C

☐

25 C

☐

0.025 C



Question 21 4.4 pts

When two point charges are 2.0 cm apart, each one experiences a 1.0-N electric force due to the other charge. If they are moved to a new separation of 8.0 cm, the electric force on each of them is closest to

hint: no math here. if the distance is x 4 (from 2 to 8), what happens to the force?

inverse square law

☐ 0.25 N.

☐ 16 N.

☐ 0.063 N.

☐ 4.0 N.

☐ 1.0 N.



Question 22 4.4 pts

Three point charges are placed on the x -axis. A charge of $+2.0 \mu\text{C}$ is placed at the origin, $-2.0 \mu\text{C}$ to the right at $x = 50 \text{ cm}$, and $+4.0 \mu\text{C}$ at the 100 cm mark. What are the magnitude and direction of the electrostatic force that acts on the charge at the origin? ($k = 1/4\pi\epsilon_0 = 9 \cdot 10^9$)

hint: call the charges q_1, q_2, q_3 . Trace the axis and place the charges with their sign.

draw the forces acting on q_1 . F_{12} and F_{13} . Find the magnitudes first. F_{13} is to the left. F_{12} to the right. So net force is right - left. Remember to convert cm to m.

☐ 0.072 N @ right

☐ 0.072 N @ left

☐ 0.05 N @ right

☐ 0.05 N @ left



Question 23 2 pts

if a glass rod is rubbed with silk, what is true

☐ silk is negative and glass positive

☐ glass is negative and silk positive



glass and silk remain neutral



glass and silk are negative



Question 24 2 pts

When I rubbed a balloon against fur,



electrons are transferred to the balloon but stay where they are placed



electrons are transferred to the balloon but spread out over the balloon



electrons are transferred to the fur and spread out over the fur



nothing happens



Question 25 2 pts

When electrons are transferred on a metallic sphere (like a van graaf dome)



they spread out over the dome



they stay where they were placed



they clump together inside the sphere, in a corner



electrons can not be transferred to a conductor



Question 26 2 pts

today the transistors in integrated circuits that acts like switches (on/off) have replaced



cathode ray tubes (CRT)



electroscopes



capacitors



leyden jars



Question 27 1 pts

A leyden jar is the ancestor of the



capacitor



resistor



transistor



tyrannosaur



Question 28 4 pts

If you rub silk with a glass rod. The silk becomes



negatively charges



positive charged



l stays neutral



too tired to check my notes



Question 29 4 pts

You rub a (party) balloon against a piece of wool (or hair). The wool (hair) becomes



positively charged.



negatively charged



stays neutral



It depends on the mass of the balloon

Quiz saved at 3:07pm

Submit Quiz